Precision@k Is a Great Target and a Dangerous Judge: Model Selection for Fixed-Budget Top-k Problems

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Abstract

In many operational problems — energy-fraud inspection, tax audits, retention campaigns, medical call-backs — a model ranks a population and a team acts only on the top k, where k is a fixed external budget. The natural objective is *precision@k*. It is well known that precision@k cannot serve as a training loss for gradient-based learners, because it is piecewise-constant and non-differentiable (and, for tree ensembles, because it does not decompose into per-node gains); we make the mechanism concrete with a worked gradient-boosting step. We then ask a less-examined question: should precision@k be the metric used for *model selection*? Across 93 imbalanced datasets (90 real, 3 synthetic) with the budget K swept over almost two orders of magnitude, and across three model families (gradient boosting, random forests, logistic regression), we find that selecting a model by validation *average precision* yields *significantly better* held-out precision@k than selecting by precision@k itself (paired Wilcoxon, $p < 1.2\times10^{-3}$ in every family): optimizing selection for the exact target metric is a losing move, a manifestation of selection-induced optimism (the winner's curse). Log-loss is an equally good selector for gradient boosting but its edge is model-dependent, whereas average precision is robust; ROC-AUC is never better than precision@k. We further show that selection difficulty is governed *not* by any scale-free ratio such as n_{pos}/K (Spearman $|\rho|\approx0.19$), but by the absolute *count* of true positives that land in the budget ($|\rho|\approx0.43$); its model-free proxy $\min(K, n_{\text{pos}})$, knowable before training, tracks that count almost perfectly ($\rho=0.94$). We give practical recommendations and release all code.

1 Introduction

Consider a utility's fraud unit whose crew can inspect a fixed number of installations per month. A meter ranked below the cutoff is never inspected, so calibration or ranking quality in the bulk of the distribution is operationally irrelevant; only the composition of the top k matters. We call this a *fixed-budget top-k problem*: score a population, act on the top k, where k is imposed from outside the model (crew size, budget, legal quota). The natural success metric is

$$\text{precision@}k = \frac{\#\{\text{true positives among the top-}k\text{ scored items}\}}{k}. \tag{1}$$

Such problems are ubiquitous (audits, marketing, triage, moderation, lead scoring).

As *background*, we first recall the well-known reason precision@k cannot be a *training* loss for gradient-based learners — it is piecewise-constant with a zero gradient — and make it concrete with a worked gradient-boosting step (Section 2). This is not a contribution; it is the setup. The

interesting question is *model selection*, where precision@k *is* admissible (we only need to compute it, not differentiate it). Here the natural instinct is compelling: since we want the model with the highest *test* precision@k, we should keep the model with the highest *validation* precision@k. Our contributions are about whether that instinct holds.

(1) Empirically, across 93 imbalanced datasets and three model families (gradient boosting, random forests, logistic regression), it does not: selecting by *average precision* yields significantly better held-out precision@k than selecting by precision@k itself, in every family (Section 5). **(2)** We explain why. Precision@k is estimated from only the k scored items in the budget, so it is a high-variance selection criterion; taking the maximum over many configurations then overfits that noise — the model-selection overfitting of Cawley and Talbot [3] and Varma and Simon [12], here specialised to a top-k metric. Average precision and log-loss instead aggregate over the *entire* validation set ($n_{\text{val}} \gg k$ points), making them far more stable estimators; that stability outweighs their looser alignment with the top-k target. **(3)** We characterise *what* governs the trade-off: not a scale-free ratio such as n_{pos}/K, but the absolute number of true positives inside the budget, whose a-priori, model-free proxy $\min(K, n_{\text{pos}})$ is knowable before training.

2 Background: why precision@k cannot be a training loss

Precision@k depends on the scores only through the *set* of top-k items. An infinitesimal change to any single score almost never changes that set, so the metric is locally constant and its gradient is zero almost everywhere; it changes only at measure-zero boundaries where two items swap across the cutoff, in jumps of $1/k$ (Fig. 1). Gradient boosting fits each new tree to the negative gradient of the loss (the pseudo-residuals); with precision@k every pseudo-residual is zero, so the ensemble never moves.

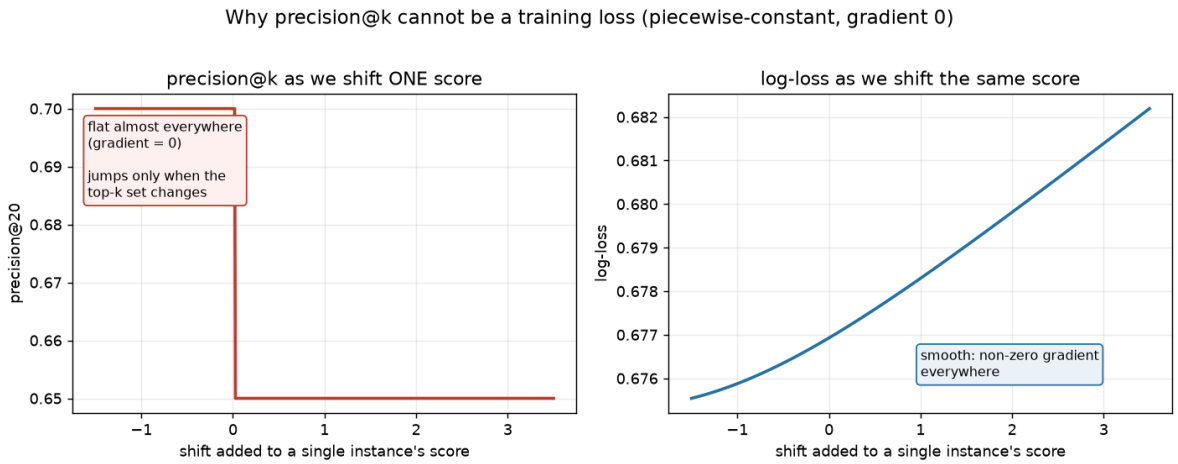

Figure 1: Sliding one item's score: precision@k is piecewise-constant with a single cliff (gradient 0 almost everywhere), whereas log-loss is smooth with an informative gradient.

Table 1 makes it concrete for $k = 3$. For log-loss with a sigmoid, the gradient w.r.t. each logit is $p - y$: false positives ranked high get large positive gradients ("push down"), missed positives get negative gradients ("push up"). Every row receives a signed instruction. Precision@3, by contrast, gives zero for every row: a helpful move (lifting D above C, changing the top-3 to $\{A, B, D\}$ and precision@3 from 1/3 to 2/3) is a *finite* jump the gradient cannot see. Differentiable surrogates for

top-k losses exist [1, 7], as does the LambdaMART device of *defining* metric-aware gradients [2]; for most systems the pragmatic recipe remains: train on log-loss, evaluate on precision@k.

The zero-gradient argument is specific to *gradient-based* learners (logistic and linear regression, neural networks, gradient boosting). Non-gradient learners such as decision trees and random forests fail to admit precision@k as a training objective for a *different* reason: they are grown by greedily optimising a *local*, per-node impurity criterion (e.g. Gini), whereas precision@k is a *global*, ranking-based, k-dependent quantity that does not decompose into per-node gains. Either way, precision@k is used only for evaluation and selection; accordingly, in our experiments no model is trained on it (gradient boosting and logistic regression minimise log-loss, random forests minimise Gini impurity).

Table 1: One boosting step, budget $k = 3$. Top-3 by score are A, B, C, so precision@3 $= 1/3$. Log-loss gradient $= p - y$; precision@3 gradient $= 0$ everywhere.

item	score p	label y	log-loss grad. $(p - y)$	precision@3 grad.
A	0.92	fraud (1)	-0.08	0
B	0.85	legit (0)	$+0.85$	0
C	0.80	legit (0)	$+0.80$	0
D	0.78	fraud (1)	-0.22	0
E	0.60	fraud (1)	-0.40	0
F	0.55	legit (0)	$+0.55$	0
G	0.30	fraud (1)	-0.70	0
H	0.10	legit (0)	$+0.10$	0

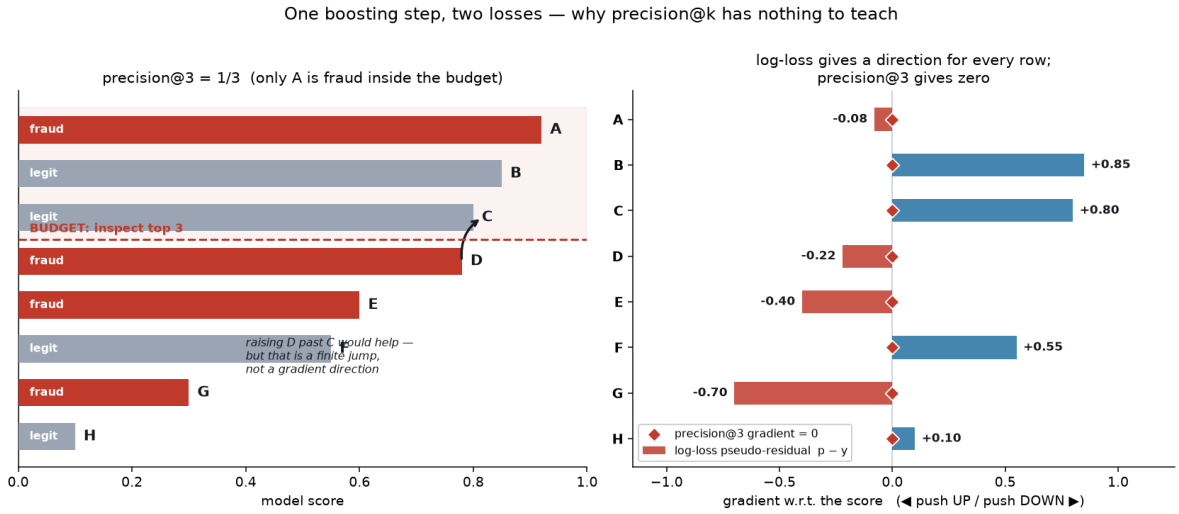

Figure 2: The same step visualised. Right: log-loss assigns a direction to every row; precision@3 assigns zero. Note log-loss already pushes C down and D up — the swap that would raise precision@3.

3 Precision@k as a selection metric: two opposing forces

Evaluation — early stopping and choosing among trained configurations — only requires *computing* the metric, so precision@k is admissible there. Two forces pull in opposite directions.

Alignment (favouring precision@k). Log-loss is a global objective: it rewards calibration across the whole distribution, not the top-k composition. A model can lower average log-loss by improving many mid-ranked cases while letting a few clear positives slip out of the budget — better by log-loss, worse at the operational objective. Only precision@k detects this.

Stability (against precision@k). Precision@k is a proportion estimated from only the k items in the budget, so its standard error scales like $\sqrt{p(1-p)/k}$ — it is a high-variance criterion. Selecting the maximum over many configurations on one validation set then partly selects noise — the winner's curse [3] — so the chosen model generalises worse and the reported number is optimistically biased. Average precision and log-loss are, by contrast, aggregates over the *whole* validation set: average precision integrates precision along the entire precision–recall curve [4], and log-loss averages a per-example term over all $n_{\text{val}} \gg k$ points. Their estimation error therefore shrinks with n_{val}, not with the tiny k, so they are far more stable selection criteria — less aligned with the exact top-k target, but much less prone to the winner's curse. Which force dominates is an empirical question.

4 Experimental protocol

Datasets. 90 real binary imbalanced datasets — the `imbalanced-learn` benchmark [9] (21 of its 27 datasets pass our minimum-positives filter) plus a de-duplicated pull from OpenML [11] (fraud, credit default, software-defect, churn, bankruptcy, medical screening, satellite/pulsar anomalies), positive rates from $\approx 20\%$ down to 0.7% — plus 3 synthetic datasets. Large sets are subsampled to 30k rows.

Model bank. Per dataset we split 50/25/25 (train/validation/test, stratified) and train a bank of C configurations with random hyperparameters. The main study uses $C = 40$ LightGBM [8] configurations (trained on log-loss); to show the conclusion is not learner-specific, Section 5 repeats the *entire* protocol with two further families — random forests and L_2-regularised logistic regression, $C = 25$ each (scikit-learn [10]). Each learner is trained with its own standard objective (log-loss for boosting and logistic regression, Gini impurity for random forests); precision@k is never a training target, only an evaluation/selection metric.

Budget sweep. Throughout, n_{pos} denotes the number of positives in the *evaluation cohort* being ranked (the test split), *not* the whole dataset — in the fraud analogy, the test split is the monthly batch of installations, n_{pos} is the frauds present in that batch, and K is how many of them you can inspect. We set K to hit target ratios $n_{\text{pos}}/K \in \{0.25, 0.5, 1, 2, 4, 8, 16\}$: a value of 16 means the cohort contains $16\times$ more positives than the budget (a tiny budget — the typical fraud regime), whereas 0.25 means the budget is four times larger than *all* the positives. This spans budgets far larger and far smaller than the positive pool. Note this ratio is distinct from the *positives that land inside* the budget (which is at most K and drives the results below); n_{pos}/K compares the budget to the whole cohort's positive pool.

Why sweep the budget? n_{pos}/K is a budget-relative measure of imbalance (= prevalence $\div$ inspection fraction), and precision@k's reliability as a *selector* is pulled by two opposing forces as K varies. (i) As an estimator it is a proportion over K items, so a larger K (smaller n_{pos}/K) lowers

its sampling variance and should make it more stable. (ii) But as K approaches the whole cohort, precision@k collapses to the base prevalence for *every* model — inspect everything and the hit rate is just the prevalence, independent of the ranking — so it loses all power to tell models apart. Precision@k can therefore only be a useful selector in a middle band, and we sweep n_{pos}/K over almost two orders of magnitude (0.25 to 16) to map it. A natural hypothesis is that this scale-free ratio predicts where precision@k is reliable; Section 5 shows it does not — what governs reliability is the absolute count of positives inside the budget, not the ratio.

Selection under noise. We bootstrap [6] the validation set 120 times; for each metric $m \in \{\text{precision@}k, \text{average precision (AP) [4]}, \text{ROC-AUC}, \text{log-loss}\}$ we select $\arg\max_c m$ and record that configuration's *true* quality, precision@k on the held-out test pool.

Normalized regret. To pool heterogeneous datasets we report, per (dataset, K),

$$\text{regret}(m) = \frac{q^\star - \bar{q}_m}{q^\star - \bar{q}_{\text{avg}}}, \tag{2}$$

where $q^\star$ is the best test precision@k in the bank (oracle), $\bar{q}_m$ the mean test quality selected by m, and $\bar{q}_{\text{avg}}$ the average model's quality. 0 means oracle selection, 1 means no better than picking at random. We keep the 322 (of 625) cases with a non-trivial gap ($q^\star - \bar{q}_{\text{avg}} \geq 0.02$). Following Demšar [5] we compare metrics with the paired Wilcoxon signed-rank test.

5 Results

Log-loss and AP beat precision@k, significantly. Table 2 and Fig. 3 summarise the pooled comparison. Log-loss and AP tie for the lowest mean regret and are statistically indistinguishable from each other ($p = 0.53$); each beats precision@k with $p < 2 \times 10^{-3}$. AUC is no better than precision@k ($p = 0.85$). In words: *selecting on the metric you ultimately care about gives you a worse model, on that same metric, than selecting on a more stable proxy.*

Table 2: Model-selection metrics pooled over 322 (dataset, budget) cases. Mean normalized regret (lower is better), win-rate (fraction of cases where the metric had the lowest regret of the four selectors, so the column sums to 100%), and paired Wilcoxon p-value versus precision@k.

selection metric	mean regret ↓	win-rate	Wilcoxon p vs. P@k
log-loss	**0.714**	**34%**	6.1×10^{-4}
average precision	0.719	21%	1.2×10^{-3}
ROC-AUC	0.770	20%	0.85 (n.s.)
precision@k	0.790	25%	—

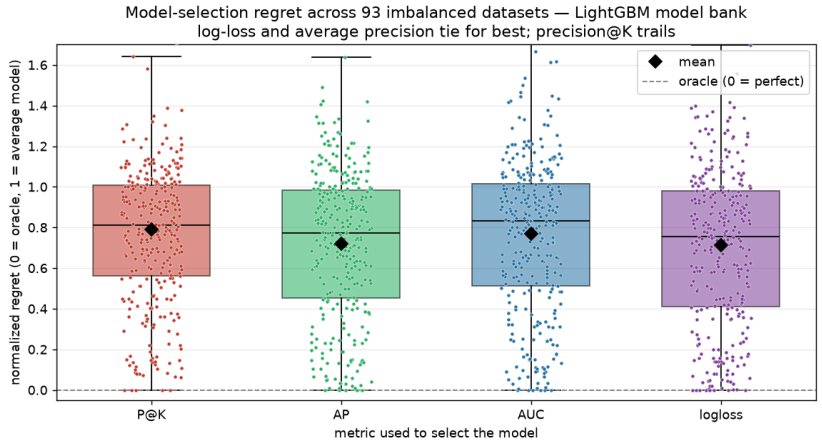

Figure 3: Distribution of normalized regret per selection metric across the 322 cases; diamonds mark means. Log-loss and average precision lead; precision@k trails.

It is a count, not a ratio. One might expect precision@k to be safe whenever the budget is large relative to the positives (large n_{pos}/K). It is not: that ratio barely orders regret (Spearman $|\rho| = 0.19$). What orders it is the *absolute number of true positives inside the budget* ($\approx$ precision $\times K$; $|\rho| = 0.43$), Fig. 4. A proportion's reliability is set by the number of successes behind it — a count, not a fraction — so two problems with the same ratio behave differently if one is ten times larger.

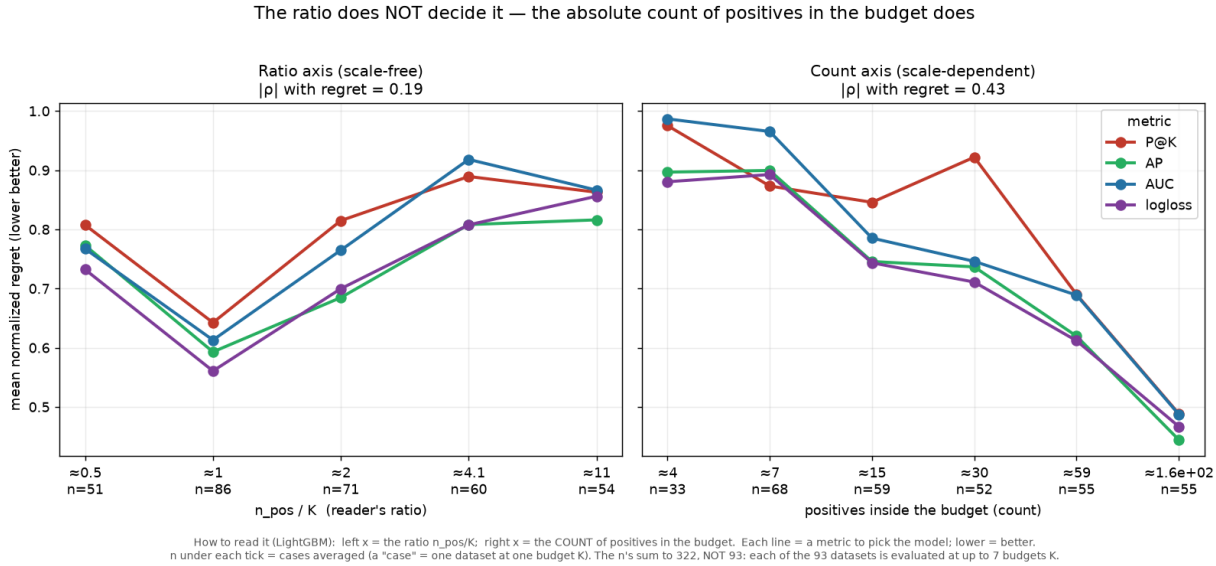

Figure 4: Left: the scale-free ratio n_{pos}/K barely orders regret. Right: the absolute count of positives inside the budget does. n = number of (dataset, budget) cases per bin.

The count is knowable a priori. The achieved count (precision $\times K$) depends on the model, but its model-free upper bound $\min(K, n_{\text{pos}})$ — the most positives that could possibly land in the budget — predicts almost as well ($|\rho| = 0.38$) and tracks the achieved count almost perfectly

($\rho = 0.94$), Fig. 5. So the deciding quantity is available before training; for the exact value, count the true positives in a single cheap baseline's top-k.

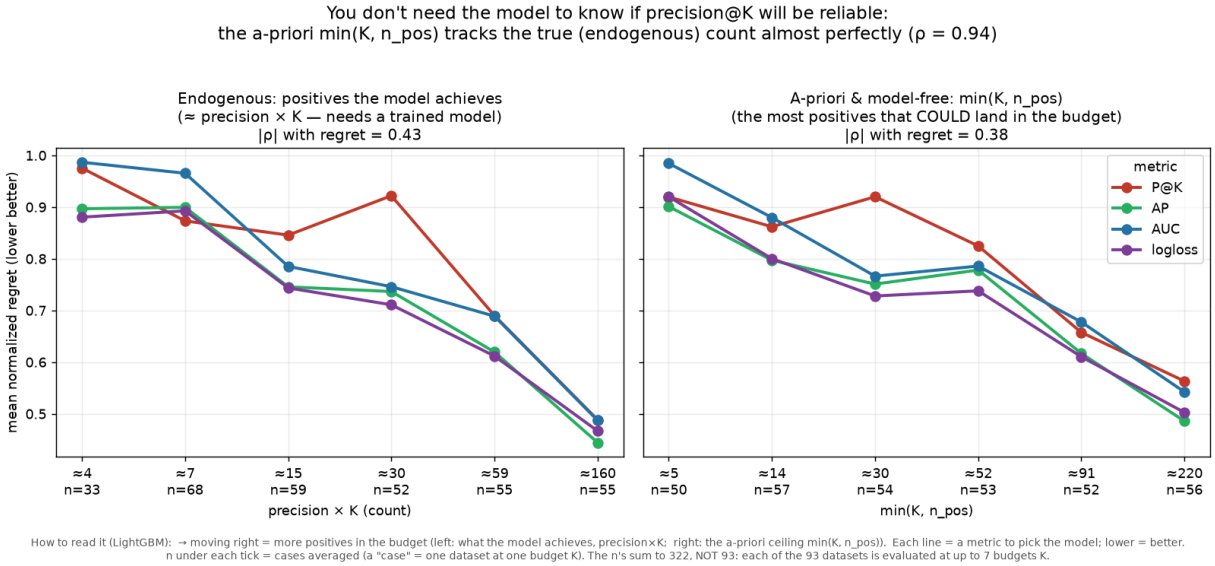

Figure 5: The endogenous count precision $\times K$ (left) and its a-priori, model-free proxy $\min(K, n_{\text{pos}})$ (right) order regret almost identically ($\rho = 0.94$ between them).

The reported number is inflated too. Consistent with the winner's curse, validation precision@k overstates true precision in proportion to how few positives sit in the budget — by tens of percent, occasionally more than 2× — so it is also an optimistic estimate of production performance (Fig. 6).

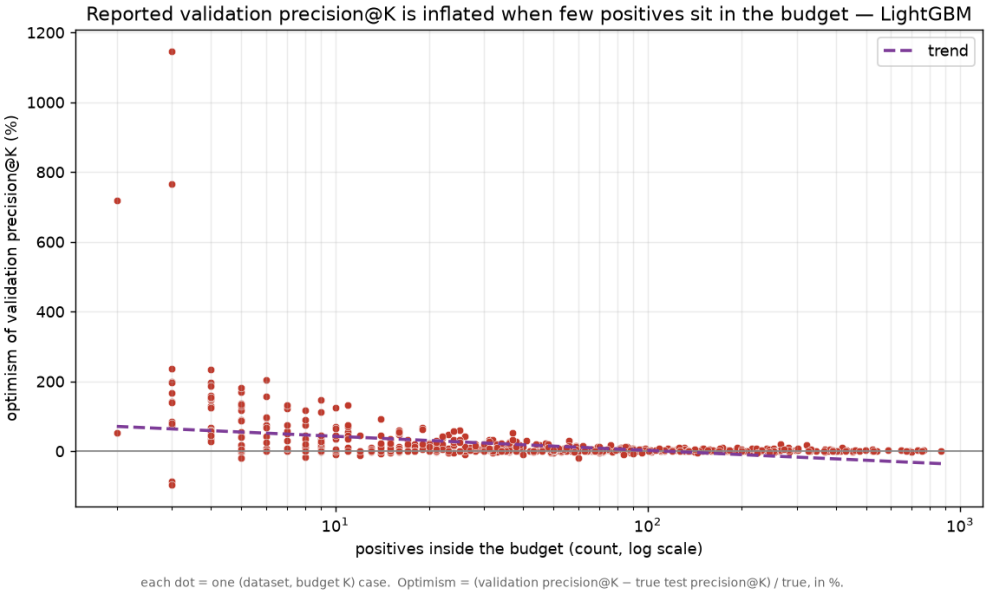

Figure 6: Optimism of validation precision@k (over-statement of the selected model's true precision) versus the number of positives in the budget.

The result holds across model families. The experiments above use LightGBM. To check that the conclusion is not an artefact of one learner, we repeat the entire selection study with two very different families: random forests and L_2-regularised logistic regression (25 configurations each, 100 bootstrap resamples, datasets subsampled to 15k rows). Table 3 and Fig. 7 show the same qualitative picture in all three families: **average precision selects significantly better models than precision@k everywhere** ($p \leq 1.2 \times 10^{-3}$), while ROC-AUC is never better than precision@k. Log-loss is tied-best for gradient boosting but does *not* beat precision@k for the other two families — its advantage is model-dependent, whereas average precision's is robust.

Table 3: Robustness across model families. Mean normalized regret (lower is better) and paired Wilcoxon p-value versus precision@k. Average precision is significantly better than precision@k in every family; log-loss only for LightGBM; ROC-AUC never.

	precision@k	average precision	log-loss	ROC-AUC
LightGBM (n=322)	0.790	0.719 (p=1.2×10^{-3})	**0.714** (6.1×10^{-4})	0.770 (n.s.)
Random forest (n=329)	0.710	**0.663** (5.7×10^{-4})	0.746 (n.s.)	0.717 (n.s.)
Logistic reg. (n=288)	0.750	**0.631** (2.9×10^{-4})	0.697 (n.s.)	0.839 (worse)

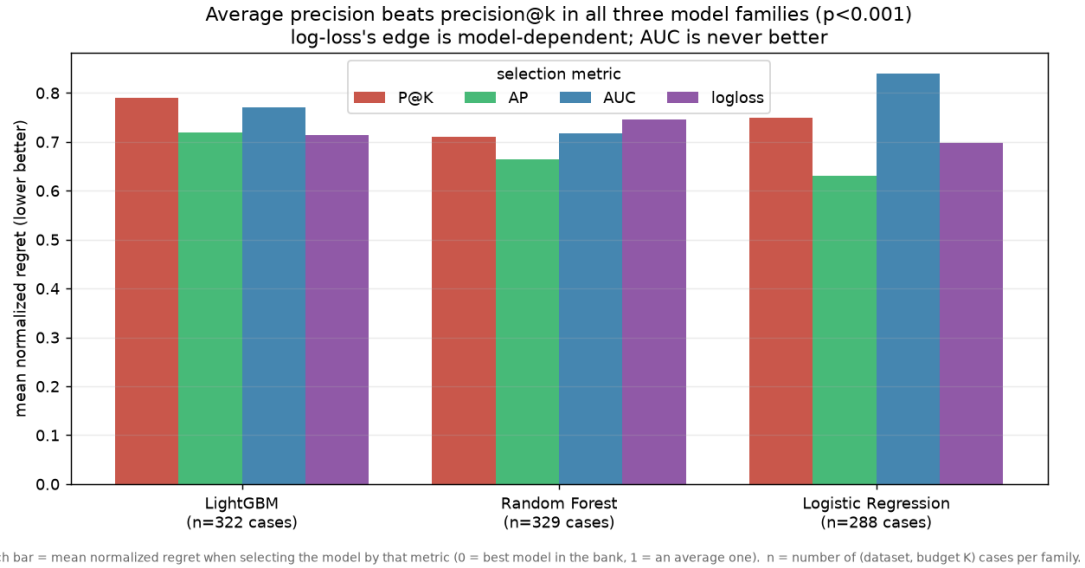

Figure 7: Mean normalized regret per selection metric, grouped by model family. Average precision (green) is lowest in all three; log-loss (purple) leads only for gradient boosting.

The count-driver result also transfers: Fig. 8 reproduces, for each family, the fall of regret with the number of positives inside the budget, with average precision at or below precision@k in nearly every bin (the sole exception is the most positive-poor bin of the random-forest panel).

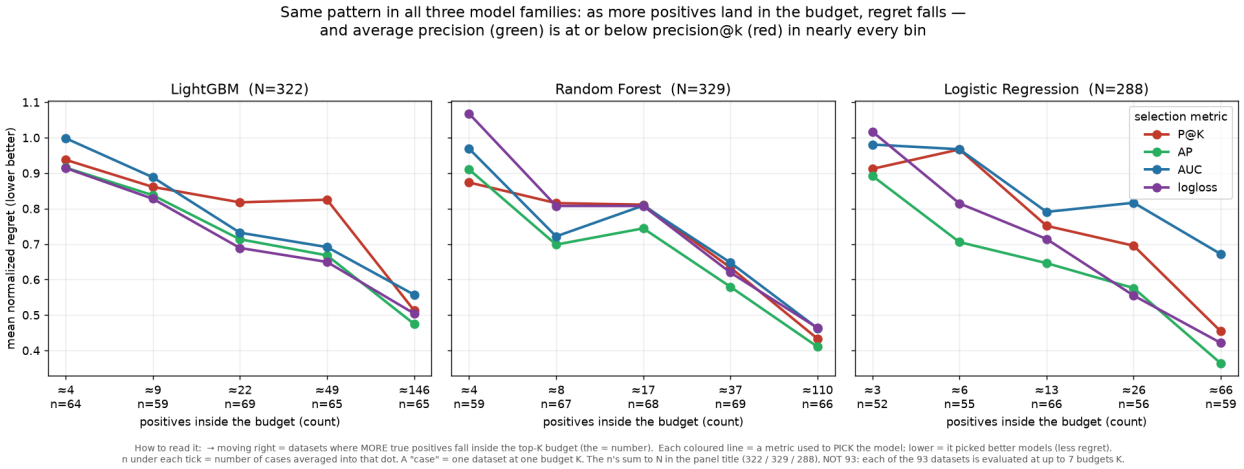

Figure 8: Regret per selection metric versus the number of true positives inside the budget, one panel per model family. In all three, regret falls as the count grows and average precision (green) is at or below precision@k (red) in nearly every bin.

The two other pieces of the fine analysis also replicate across families. The scale-free ratio n_{pos}/K fails to order regret in *any* family (Fig. 9: the curves are non-monotone and tangled), whereas along the a-priori, model-free proxy $\min(K, n_{\text{pos}})$ regret falls cleanly in all three (Fig. 10; monotone for the stable metrics, with a small bump in the precision@k curve). The count of positives in the budget — not a ratio — is what governs selection difficulty, regardless of the learner.

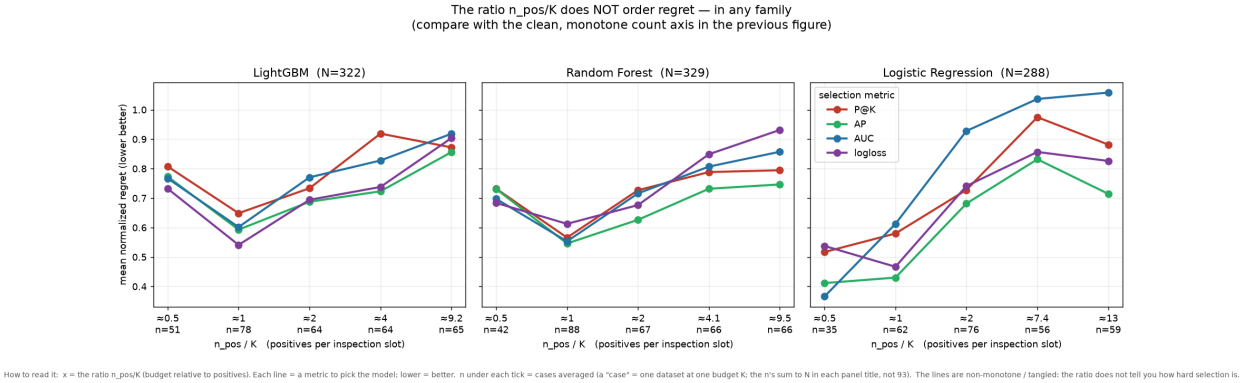

Figure 9: Regret versus the ratio n_{pos}/K, one panel per family. Unlike the count axis, the ratio does not order regret in any family — confirming that a scale-free ratio is the wrong lens.

Figure 10: Regret versus the a-priori proxy $\min(K, n_{\text{pos}})$, one panel per family. Knowable before training, regret falls with it in all three families, with average precision (green) at or below precision@k (red) in every bin.

6 Discussion and recommendations

Keep precision@k as the target and the headline evaluation metric, but not as the training loss and not, by default, as the selection metric. For selection, **default to average precision** — it was the only metric to beat precision@k significantly in every model family; log-loss is an equally good choice for gradient boosting but not in general. Use precision@k only as a tie-breaker. The single lever that matters is knowable in advance: $\min(K, n_{\text{pos}})$, the number of positives that can land in the budget. When it is small, no selection metric is reliable — invest in more labelled positives or a larger budget, not a cleverer metric. Finally, never quote raw validation precision@k as expected production performance when the budget is positive-poor; report a nested/held-out estimate.

7 Is there an optimal way to select from these metrics?

We have seen that average precision (and, for boosting, log-loss) selects better models than precision@k even though precision@k is the deployment objective. Can we do better still by *combining* the four validation metrics? This section searches for such a rule and, more importantly, for a *ceiling* on how well any rule can do. It is a separate, smaller-scale analysis ($C = 30$ configurations, datasets subsampled to 15k, metrics on the full validation set, no bootstrap), so its absolute regrets are *not* comparable to the tables above — only comparisons *within* it are meaningful. *Caveat:* the weighting and cascade tune a choice on the test set, so their regret numbers are optimistic; read them as qualitative recipes.

A ceiling: the winner is hard to find from validation. Before combining metrics, ask how well any of them *could* work. For each case we locate the configuration with the best *test* precision@k (the oracle) and record its percentile in each *validation* ranking (Fig. 11). The answer is sobering: the test-winner sits only in the mid-to-upper range of every validation metric — median percentile ≈ 67 (precision@k), 68 (AP), 68 (AUC), 58 (log-loss) — and is widely spread, frequently landing below the median. *No* validation metric reliably puts the true winner near the top; average precision and AUC do so marginally better than precision@k and log-loss, consistent with the earlier results, but the effect is small. This is why selection regret plateaus around the same level whatever metric or combination we use: the information simply is not there in a single validation split.

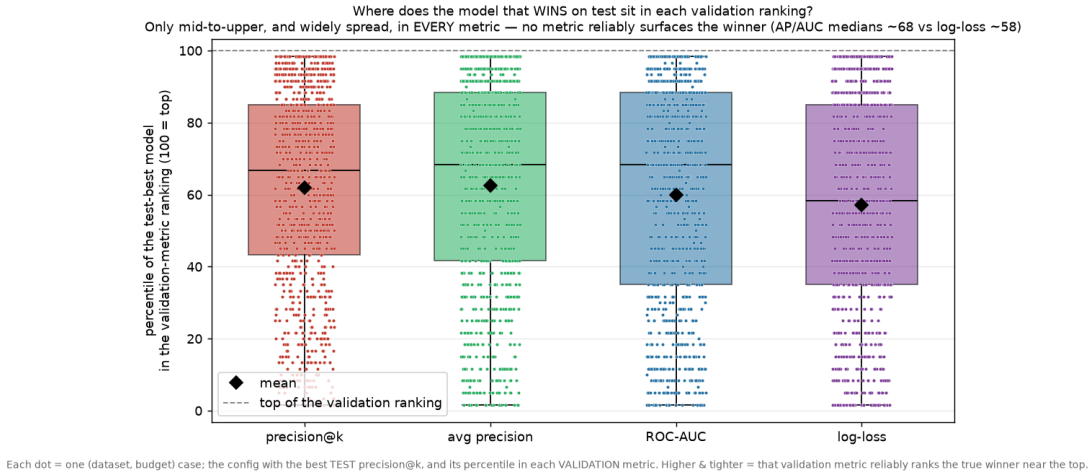

Figure 11: Percentile of the test-best model within each validation-metric ranking, across all cases (100 = top of the validation ranking). The winner is only mid-to-upper and widely spread in every metric — a ceiling on what any selection rule can achieve.

How much should each metric count? We standardise the four validation metrics within each case and select the configuration maximising a weighted sum $\sum_m w_m z(m)$, then search the weight simplex for the w minimising mean test regret (Fig. 12). One robust fact emerges: a *blend* beats every single metric in all three families (e.g. pooled regret 0.64 vs. 0.66 for the best solo metric). *The individual weights, however, are not identifiable and should not be read literally.* The four metrics are strongly correlated (pairwise Pearson 0.45–0.73 across configurations — they all reward ranking positives high), so they are near-substitutes and the optimal weight can shift between them with almost no effect; and the regret surface is nearly flat — for LightGBM, 11 of 286 grid weightings lie within 0.01 of the optimum, ranging from AUC-heavy $[0.3, 0.1, 0.5, 0.1]$ to precision@k-heavy $[0.8, 0, 0.1, 0.1]$ to log-loss-heavy $[0.2, 0, 0.3, 0.5]$. So the per-family split in Fig. 12 is largely test-fit noise. The one defensible reading is qualitative: do not rely on precision@k alone; a combination of stable ranking metrics is marginally better, and (next) a simple cascade captures that robustly.

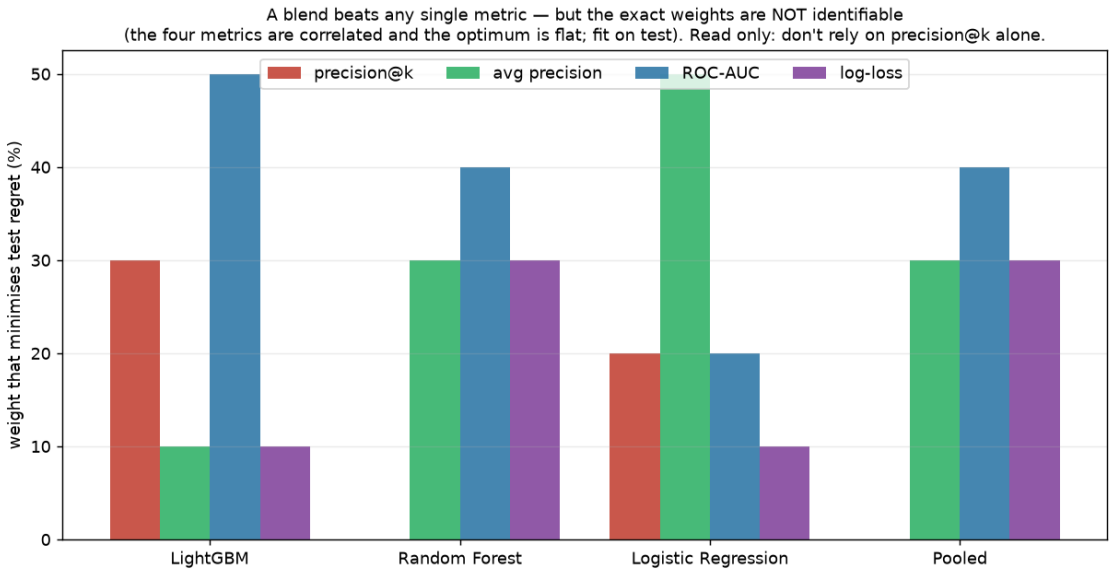

Figure 12: Weights on the four validation metrics that minimise test regret, per family (and pooled). Fit on test and *not identifiable* — the metrics are correlated and the optimum is flat, so the exact split is largely noise (see text). The only robust reading: a blend beats any single metric and precision@k is not the answer alone.

A robust cascade. A safer, almost parameter-free strategy uses precision@k the way our results suggest — as a *tie-breaker*, not a primary selector: shortlist the top-n configurations by validation average precision, then keep the one with the best validation precision@k (Fig. 13). Here n=1 is pure average precision and large n lets precision@k decide. A small shortlist ($n \approx 3$) is at or below pure average precision and below the precision@k-dominated end in every family (pooled: 0.64 at n=3 vs. 0.66 for pure average precision at n=1 and 0.67 at n=30). A relevant subtlety: precision@k takes only discrete values (multiples of $1/K$), so many configurations tie on it — selecting by precision@k alone is under-determined, and *how* you break those ties matters (breaking them arbitrarily is worse still). The cascade resolves them with the stable metric, which is exactly why it helps. This quantifies the practical rule: **select with average precision, and let precision@k only break ties among already-strong models**.

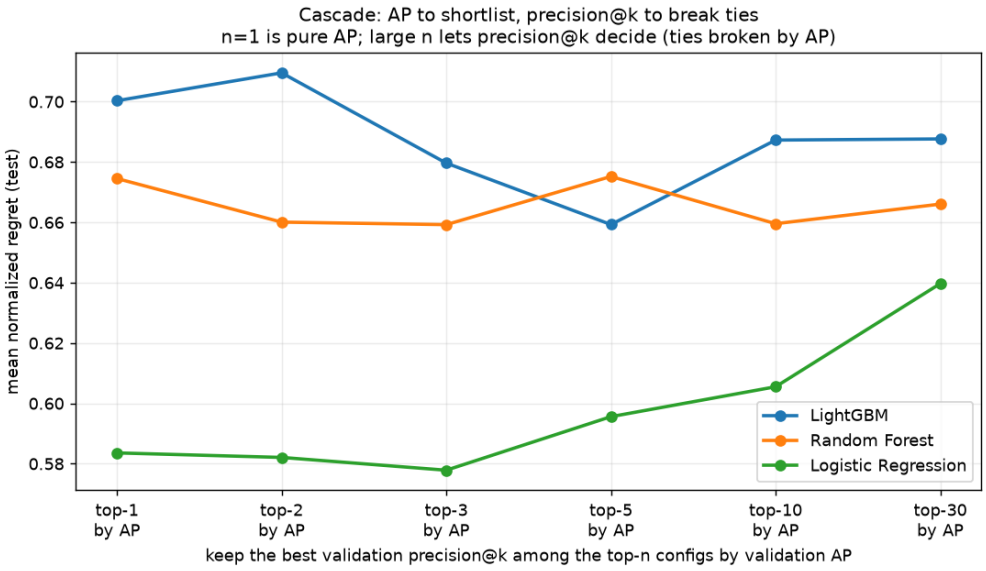

Figure 13: Cascade strategy: shortlist by validation average precision, break ties by validation precision@k. A shortlist of $n \approx 3$ beats both the precision@k-dominated end (n=30) and pure average precision (n=1) in all three families.

Do fancier combinations help? Barely. We also tried rank-based combinations (selecting the configuration with the best sum, or the best minimum, of validation percentile ranks across metrics). Table 4 collects them. Two conclusions. First, any rule that *combines average precision with precision@k* — the top-3 cascade, or the AP+precision@k rank-sum — edges out pure average precision and clearly beats pure precision@k, so there is a mild "sweet spot" in being high on *both*. Second, the gains are small and no rule wins in every family (rank-min is best for random forests, the cascade for logistic regression, the AP+precision@k rank-sum for LightGBM), exactly as the ceiling in Fig. 11 predicts. The practical takeaway is unchanged and robust: **rank by average precision, use precision@k to break ties among the strongest few** — and don't expect a magic combination, because the winner is not identifiable from one validation split.

Table 4: Mean normalized regret (lower better) of selection rules built from the four validation metrics. Combining average precision with precision@k (cascade or rank-sum) is the sweet spot, but improvements over pure average precision are small. Test-tuned, so optimistic.

selection rule	LightGBM	random forest	logistic reg.	pooled
pure precision@k	0.709	0.670	0.711	0.696
pure average precision	0.700	0.675	0.584	0.656
cascade: top-3 AP $\rightarrow$ P@k	0.680	0.659	**0.578**	**0.641**
rank-sum AP + P@k	**0.663**	0.671	0.616	0.652
rank-min(AP, P@k)	0.673	**0.655**	0.655	0.661

8 Limitations

We study three model families (gradient boosting, random forests, logistic regression) but a fixed model bank, isolating selection noise but not early stopping (where the same variance argument

applies). Ground-truth precision@k on small datasets is itself estimated with noise, mitigated by pooling and by robust statistics (rank, win-rate, signed-rank tests). The Wilcoxon tests treat the (dataset, K) cases as independent, but the up-to-seven budgets per dataset are correlated, so the p-values are likely anti-conservative; the effect directions are consistent across budgets and families, but a dataset-level aggregation would firm up the significance. Results are statements about *expected* selected quality. Extending to deep models, and to explicit top-k surrogate training, is future work.

9 Conclusion

The metric you optimize and the metric you select with need not be the same. For fixed-budget top-k problems *on imbalanced data* — the regime studied here, and the one in which these problems arise — selecting on *average precision* robustly beats selecting on the noisy target metric itself, across 93 imbalanced datasets and three model families, and the deciding factor for how much any of this matters is a plain count, how many positives your budget will contain, not a ratio of rates. Whether the effect persists on near-balanced data, where the top-k slice is less positive-starved, is left to future work.

Reproducibility. All code, the dataset list, and figures are released at `https://github.com/alejandrogomez97/precision-at-k-model-selection` and run from a fixed seed.

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